NAVRACHANA UNIVERSITY

Teaching Plan

Computer Vision I (CSE615)

Credit : 2

<table>
<tr><td>Faculty</td><td>
<table>
<tr><td>Name</td><td>Contact</td><td>Office Hours</td><td>Office Location</td></tr>
<tr><td>Ashish Jani</td><td>ashish.jani@nuv.ac.in</td><td></td><td></td></tr>
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<tr><td>School</td><td>SET</td></tr>
<tr><td>Semester</td><td>Spring (Jan to June 2026)</td></tr>
<tr><td>Lecture time & Weedays</td><td>
<table>
<tr><td rowspan="2">SECTION_1</td><td>Thursday</td><td>10:00:00-11:00:00</td><td>05-01-2026 TO 08-05-2026</td></tr>
<tr><td>Friday</td><td>10:00:00-11:00:00</td><td>05-01-2026 TO 08-05-2026</td></tr>
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<tr><td>Pre-requisite</td><td></td></tr>
<tr><td>Co-requisite</td><td></td></tr>
<tr><td>Anti-requisite</td><td></td></tr>
<tr><td>Course Description</td><td>This course enables Computer Vision focuses on development of algorithms and techniques to analyze and interpret the visible world around us.</td></tr>
<tr><td>Course Abstract</td><td>Understanding of the fundamental concepts related to multi-dimensional signal processing, feature extraction, pattern analysis visual geometric modelling, stochastic optimization etc. Knowledge of these concepts is necessary in this field, to explore and contribute to research and further developments in the field of computer vision. Applications range from Biometrics, Medical diagnosis, document processing, mining of visual content, to surveillance, advanced rendering etc.</td></tr>
<tr><td>Course Objectives</td><td>1. To teach theoretical concepts underlying computer algorithm development.
2. To teach problem solving by different methods.
3. To Synthesize efficient algorithms in common engineering design situations.
4. To teach analyzing algorithms for time and space complexity through different methods such as statistical methods and dynamic programming.
5. To develop ability to understand how the choice of data structures and the algorithm design methods impact the performance of programs.
6. To identify the key characteristics of a given problem and analyze the suitability of a specific algorithm design technique for the problem.</td></tr>
<tr><td>Course Outcomes</td><td>Student will be able to:
1. To implement fundamental image processing techniques required for computer vision
2. Understand Image formation process
3. To perform shape analysis
4. Extract features form Images and do analysis of Images
5. Generate 3D model from images
6. To develop applications using computer vision techniques
7. Understand video processing, motion computation and 3D vision and geometry</td></tr>
<tr><td>Course Typology</td><td>Theory</td></tr>
<tr><td>Course Outline</td><td>The course contents are divided in to 5 units. The topic covers Basic principles of image formation, image processing algorithms and different algorithms for 3D reconstruction and recognition from single or multiple images (video). This course emphasizes the core vision tasks of scene understanding and recognition. Applications to 3D modeling, video analysis, video surveillance, object recognition and vision based control will be discussed.</td></tr>
<tr><td>Pedagogy</td><td>Theory, Discussions, Demonstration of Applications, and Practical/Hands-on and Project based Learning.</td></tr>
<tr><td>Expectations from Students</td><td>1) Maintain silence in class.
2) Students should attend 100% classes.
3) Don’t distract or harass others
4) Participate in class discussions
5) Turn in and do well in Assignments/Homework
6) Demonstrate in class presentation/discussions
7) Pass in Midterm and Final exam
8) Arrive to class on time
9) Maintain 80% attendance (minimum)
10) No plagiarism at all
11) Come prepared to class</td></tr>
<tr><td>Assessment / Evaluation</td><td></td></tr>
<tr><td>Project / Assignment Details</td><td>Students will do one group project covering this course. A group of maximum 2 students is allowed. At the end they will present their work.</td></tr>
<tr><td>Course Material</td><td>-</td></tr>
<tr><td>Additional Information</td><td>--</td></tr>
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Session Plan

Topic Title	Session No.	Topic & Subtopic Details	Readings, Cases, etc.	Activities	Important Dates
Image Processing, Computer Vision and Computer Graphics , What is Computer Vision	1	Image Processing, Computer Vision and Computer Graphics , What is Computer Vision	Computer Vision -A modern approach	Task/Project	
-Low-level, Mid-level, High-level , Overview of Diverse Computer Vision Applications:	2	-Low-level, Mid-level, High-level , Overview of Diverse Computer Vision Applications:	Computer Vision -A modern approach	Task/Project	
Document Image Analysis, Biometrics, Object Recognition, Tracking,	3	Document Image Analysis, Biometrics, Object Recognition, Tracking,	Computer Vision -A modern approach	Task/Project	
Medical Image Analysis, Content-Based Image Retrieval, Video Data Processing,	4	Medical Image Analysis, Content-Based Image Retrieval, Video Data Processing,	Computer Vision -A modern approach	Task/Project	
Virtual Reality and Augmented Reality.	5	Virtual Reality and Augmented Reality.	Computer Vision -A modern approach	Task/Project	
Monocular imaging system , Radiosity:	6	Monocular imaging system , Radiosity:	Computer Vision -A modern approach	Task/Project	
	7		Computer Vision -A modern approach	Task/Project	
Radiance, Irradiance,	8	Radiance, Irradiance,	Computer Vision -A modern approach	Task/Project	
BRDF, color etc,	9	BRDF, color etc,	Computer Vision -A modern approach	Task/Project	
Orthographic	10	Orthographic	Computer Vision -A modern approach	Task/Project	
Perspective Projection,	11	Perspective Projection,	Computer Vision -A modern approach	Task/Project	
Camera model and Camera calibration,	12	Camera model and Camera calibration,	Computer Vision -A modern approach	Task/Project	
Binocular imaging systems, Multiple views geometry,	13	Binocular imaging systems, Multiple views geometry,	Computer Vision -A modern approach	Task/Project	
Structure determination,	14	Structure determination,	Computer Vision -A modern approach	Task/Project	
shape from shading, Photometric Stereo, Depth from Defocus	15	shape from shading, Photometric Stereo, Depth from Defocus	Computer Vision -A modern approach	Task/Project	
Construction of 3D model from images	16	Construction of 3D model from images	Computer Vision -A modern approach	Task/Project	
Construction of 3D model from images	17	Construction of 3D model from images	Computer Vision -A modern approach	Task/Project	
Edges - Canny, LOG, DOG; Line detectors (Hough Transform),	18	Edges - Canny, LOG, DOG; Line detectors (Hough Transform),	Computer Vision -A modern approach	Task/Project	
Corners - Harris and Hessian Affine,	19	Corners - Harris and Hessian Affine,	Computer Vision -A modern approach	Task/Project	
Orientation Histogram, SIFT	20	Orientation Histogram, SIFT	Computer Vision -A modern approach	Task/Project	
SURF, HOG, GLOH	21	SURF, HOG, GLOH	Computer Vision -A modern approach	Task/Project	
Scale-Space Analysis- Image	22	Scale-Space Analysis- Image	Computer Vision -A modern approach	Task/Project	
Pyramids and Gaussian derivative filters,	23	Pyramids and Gaussian derivative filters,	Computer Vision -A modern approach	Task/Project	
Gabor Filters and DWT.	24	Gabor Filters and DWT.	Computer Vision -A modern approach	Task/Project	
Background Subtraction and Modeling,	25	Background Subtraction and Modeling,	Computer Vision -A modern approach	Task/Project	
KLT	26	KLT	Computer Vision -A modern approach	Task/Project	
Optical Flow	27	Optical Flow	Computer Vision -A modern approach	Task/Project	
Spatio -Temporal Analysis,	29	Spatio -Temporal Analysis,	Computer Vision -A modern approach	Task/Project	
Dynamic Stereo	30	Dynamic Stereo	Computer Vision -A modern approach	Task/Project	

Publish Status	Published